

Circuits

Circuit Equations

Current: $I = \frac{dQ}{dt} = n|q|v_d A$

(n = number of charges; q = charge per particle; v_d = drift velocity; A = area of conductor)

Current Density: $\vec{J} = nq\vec{v}_d = \frac{I}{A}$

Resistivity: $\rho = \frac{E}{J}$

(J = current density; E = electric field)

Resistivity: $\rho(T) = \rho_0[1 + \alpha(T - T_0)]$ (for small temperature changes)

Resistance: $R = \frac{\rho L}{A}$

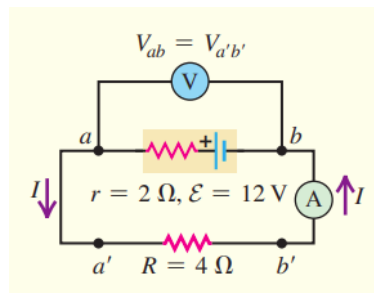
Ohm's Law: $V = IR$

Power: $P = V_{ab}I$

Power (to resistor): $P = I^2 R = \frac{V_{ab}^2}{R}$

Conduction in metals: $\rho = \frac{m}{ne^2\tau}$

Circuits



Every real source of emf has some internal resistance r , so its terminal potential difference is always

$$V_{ab} = \varepsilon - Ir$$

Resistors in Circuits

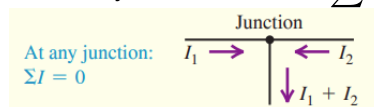
For series: $R_{eq} = R_1 + R_2 + \dots$

For parallel: $R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \dots \right)^{-1}$

Kirchhoff's Rules:

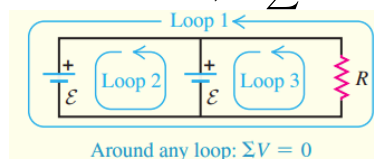
Junction/Current Rule:

At any junction, the algebraic sum of the currents into a junction is zero, i.e. $\sum I = 0$



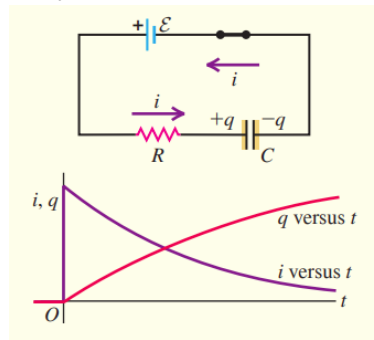
Loop/Voltage Rule:

At any loop, the algebraic sum of the potential differences is zero, i.e. $\sum V = 0$



R-C Circuits

When a capacitor is charged by a battery in series w/ a resistor, the current and capacitor charge are not constant, i.e. the charge and current approaches zero asymptotically:



Capacitor Charging:

For the charge inside the capacitor:

$$q = C\varepsilon(1 - e^{-t/RC})$$

$$= Q_f(1 - e^{-t/RC})$$

For the current flowing throughout the circuit:

$$i = \frac{dq}{dt} = \frac{\varepsilon}{R}e^{-t/RC} = I_0e^{-t/RC}$$

Capacitor Discharging:

For the charge inside the capacitor:

$$q = Q_0e^{-t/RC}$$

For the current flowing throughout the circuit:

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC}e^{-t/RC} = I_0e^{-t/RC}$$

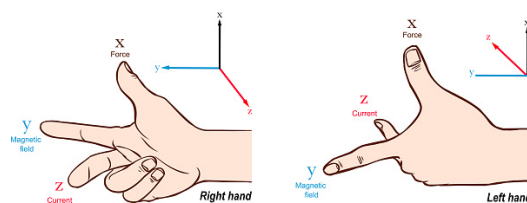
Magnetism

Magnetic Force

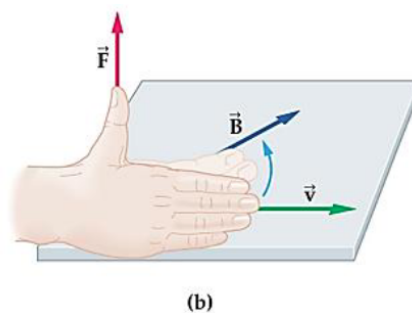
On Charged Particles

Charge needs to move in a magnetic field to experience a force. For direction, refer to Fleming's hand rules (left is for force generation, right is for induction)

Fleming's **Left** Hand Rule And Fleming's **Right** Hand Rule



Alternatively, right hand rule

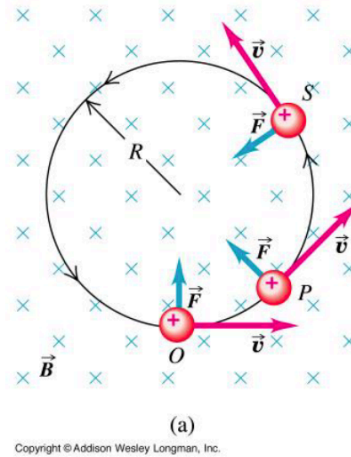


Can be found through either

$$F = |q|vB \sin \theta \quad \vec{F} = q\vec{v} \times \vec{B}$$

where B is the magnetic field, measured in Tesla (T) with SI units NS/Cm.

Because the force is perpendicular to the direction of the field, it can be said that no work is being done. A charge can have a circular path:



where

$$\begin{aligned} F_C &= m \frac{v^2}{R} = F_M \\ qvB &= m \frac{v^2}{R} \\ R &= \frac{mv}{qB} \end{aligned}$$

which results in a helical path.

On Current-carrying Conductor

$$\begin{aligned} F_M &= \Delta q v B \sin \theta \\ &= \left(\frac{\Delta q}{\Delta t} \right) (v \Delta t) B \sin \theta \\ &= I(\Delta L) B \sin \theta \end{aligned}$$

where its vector form will be

$$\vec{F} = I(\vec{L} \times \vec{B})$$

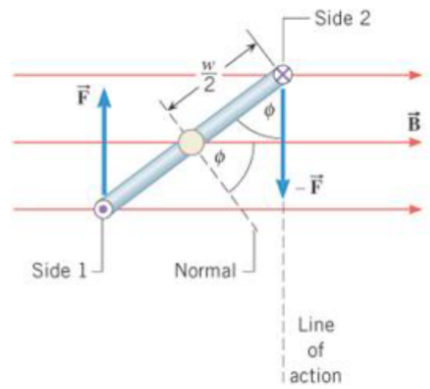
If conductor is not a straight line, divide into segments dL :

$$d\vec{F} = I(d\vec{L} \times \vec{B})$$

such that

$$\vec{F} = \int I(d\vec{L} \times \vec{B})$$

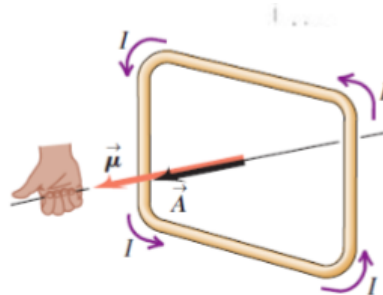
Force and Torque on a Current Loop



Net force is zero, but net torque is not zero.

$$\begin{aligned}\tau &= 2(ILB) \left(\frac{1}{2} W \sin \phi \right) \\ &= I(LW)B \sin \phi \\ &= IAB \sin \phi\end{aligned}$$

where $A = LW$ is the area of the loop.



For loop with N turns:

$$\tau = NIAB \sin \phi$$

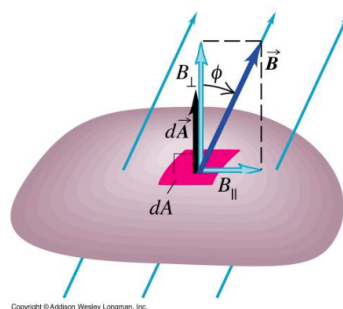
In vector form:

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

where $\mu = NIA$

Magnetic Flux

Given a field B going through a surface:



$$\Phi_B = B \perp A = BA \cos \phi$$

where $B = \frac{\Phi_B}{A}$ if B is perpendicular to the surface area. For an irregular area, this can be expressed as

$$\Phi_B = \int_S \vec{B} \cdot d\vec{A}$$

With Gauss' law, for a closed surface, this value is 0.

Applications of Motion of Charged Particles

Velocity Selector and Thomson's Experiment

$$v = \frac{E}{B} \quad \frac{e}{m} = \frac{E^2}{2VB^2}$$

Mass Spectrometer

For circular path:

$$r = \frac{mv}{qB}$$

such that the energy of particle

$$\frac{1}{2}mv^2 = qV$$
$$m = \left(\frac{qr^2}{2V} \right) B^2$$

Hall Effect

Hall voltage:

$$V_H = E_c \Delta z = v_d B_y \Delta z$$
$$= \frac{J_x B_y \Delta z}{nq}$$

Magnetic Sources

Current carrying wires can produce magnetic fields.

The force between parallel wires:

$$F = 2k_m \frac{I_1 I_2}{d} L$$

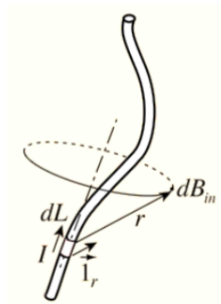
A moving point charge q:

$$B = k_m \frac{|q|v \sin \phi}{r^2} \quad \vec{B} = k_m \frac{|q|\vec{v} \times \vec{r}}{r^3}$$

where $k_m = \frac{\mu_0}{4\pi} = 1 \times 10^{-7} T \cdot m / A$

The direction is dictated by right hand rule.

Biot-Savart Law



Given the above:

$$d\vec{B} = k_m \frac{Id\vec{L} \times \vec{r}}{r^3}$$
$$\vec{B} = k_m \int \frac{Id\vec{L} \times \vec{r}}{r^3} = k_m \int \frac{Id\vec{L} \times \hat{r}}{r^2}$$

Ampere's Law

The line integral of B along a closed path is always the same.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

Because of this, in a current-carrying solenoid:

Inside the solenoid: $B_{\text{total}} = \mu_0 n I$ (where n is the number of turns)

Outside the solenoid: $B_{\text{total}} = 0$